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4. Same example, new initial conditions

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Calculator

Recitation due Mar 6, 2024 20:30 IST Completed

LTI Consumer (External resource) (0.0 / 1.0 points)

3x3 system continued

Use your code from above to solve the same problem with $a_{12} = 1, a_{13} = 1/2, a_{23} = 2$. You should
conditions

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

What do you notice about the long time behavior?

Script

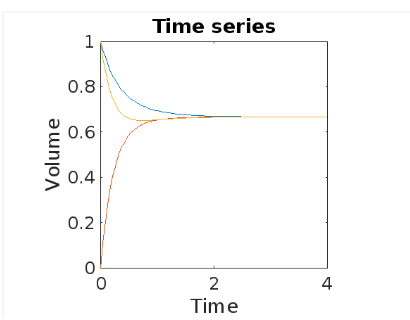
Save Reset

MATLAB Documentation (https://www.mathworks.com/help/)

```
% Input the 3x3 matrix describing your linear system as a variable A
A = [-3/2 1 1/2; 1 -3 2; 1/2 2 -5/2];
% Input your initial conditions as a column vector x0
x0 = [1;0;1];
% Use eig(A) to find the eigenvalues and eigenvectors of A
% Define the eigenvectors as column vectors v1, v2 and v3,
% and the eigenvalues as lambda1, lambda2, lambda3
% so that A*v1 = lambda1*v1, etc.
[V,D] = eig(A);
v1 = V(:,1);
v2 = V(:,2);
v3 = V(:,3);
lambda1 = D(1,1);
lambda2 = D(2,2);
lambda3 = D(3,3);
% Calculate the column vector c = [c1;c2;c3] from the initial conditions usi
c = inv(V)*x0;
c1 = c(1,1); c2 = c(2,1); c3 = c(3,1);
%%% Define a row vector t with 100 equally spaced entries,
%%% beginning with 0 and ending at 4.
t = linspace(0,4,100);
%%% Define three row vectors h1, h2 and h3, with entries corresponding
%%% to h1(t), h2(t) and h3(t) evaluated at each time in t.
h1 = c1*exp(lambda1*t)*v1(1) + c2*exp(lambda2*t)*v2(1) + c3*exp(lambda3*t)*
h2 = c1*exp(lambda1*t)*v1(2) + c2*exp(lambda2*t)*v2(2) + c3*exp(lambda3*t)*
h3 = c1*exp(lambda1*t)*v1(3) + c2*exp(lambda2*t)*v2(3) + c3*exp(lambda3*t)*
%%% Now use plot to plot the three vectors against time on the same figure
plot(t,h1)
hold on
plot(t,h2)
plot(t,h3)
set(gca,'fontsize',18)
xlabel('Time')
ylabel('Volume')
title('Time series')
axis square
```

Run Script

Output



Remark about symmetric matrices

Calculator

This is a symmetric matrix, and so it will always have real eigenvalues. In particular, the characteristic polynomial of this matrix is

$$\lambda\left(\lambda^2+2\left(a_{12}+a_{13}+a_{23}\right)\lambda+3\left(a_{12}a_{13}+a_{12}a_{23}+a_{13}a_{23}\right)\right)=0$$

and so $\mathbf{A}$ has one zero eigenvalue. The remaining quadratic always has two negative roots for $a_{ij}>0$. So for large time solutions tends towards a constant. Also note

$$\frac{d}{dt}(h_1+h_2+h_3)=0$$

so the final constant solution is determined by the initial conditions.

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4. Same example, new initial conditions

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